

Prompt PRD – pip Voice AI Assistant

Bertindaklah sebagai **Principal AI Architect, Principal Backend Engineer, Principal Mobile Engineer (iOS & Android), Voice AI Engineer**, dan **Product Manager**.

Buat **Product Requirement Document (PRD)** dan **Technical Design Document (TDD)** yang lengkap untuk fitur **Rabbit Voice AI Assistant**.

Gunakan **Bahasa Indonesia**.

Dokumen harus dapat langsung dijadikan acuan oleh tim Product, AI, Backend, iOS, Android, DevOps, dan QA.

Tujuan Produk

Rabbit Voice adalah AI Voice Assistant yang mampu melakukan percakapan dua arah secara natural seperti berbicara dengan manusia.

Target pengalaman pengguna setara dengan:

- ChatGPT Voice
- Gemini Live
- Siri
- Google Assistant

Fokus utama:

- Latency rendah
 - Streaming realtime
 - Natural conversation
 - Interruptible conversation
 - Multi language
 - Multi AI Provider
 - Offline fallback
-

Platform

Mobile:

- iOS (SwiftUI)
- Android (Jetpack Compose)

Backend:

- Python
 - FastAPI
 - WebSocket
-

Voice Features

Support:

- Push To Talk
 - Continuous Conversation
 - Full Duplex (Future)
 - Streaming Audio
 - Streaming Response
 - Voice Activity Detection (VAD)
 - Noise Suppression
 - Echo Cancellation
 - Silence Detection
 - Auto Stop Speaking
 - Interrupt AI ketika user mulai berbicara
 - Resume Conversation
 - Wake Word (Future)
-

Voice State Machine

Jelaskan state berikut beserta transisinya:

Idle

Listening

Processing

Thinking

Speaking

Interrupted

Paused

Disconnected

Error

Reconnecting

Voice Pipeline

Buat diagram lengkap.

Contoh:

Microphone

↓

Noise Suppression

↓

Echo Cancellation

↓

Voice Activity Detection

↓

Speech To Text

↓

Language Detection

↓

Conversation Manager

↓

Prompt Builder

↓

Memory



AI Router



LLM



Text To Speech



Streaming Audio



Speaker

AI Router

Gunakan:

Strategy Pattern

Factory Pattern

Provider Registry

Jangan menggunakan if/else.

Provider dapat diganti melalui konfigurasi.

AI Provider

Support:

OpenAI

Google Gemini

Claude

Groq

OpenRouter

Ollama

DeepSeek

Mistral

xAI

Jelaskan:

- kapan provider dipilih
- priority
- fallback
- retry
- timeout
- health check

Voice Provider

Support:

OpenAI Realtime

Gemini Live

ElevenLabs

System Voice

Piper

Kokoro TTS

Whisper

Google Speech

Deepgram

Azure Speech

Jelaskan fungsi masing-masing.

Speech To Text

Bandingkan:

Whisper

Google Speech

Deepgram

Azure

OpenAI Speech

Gemini Live

Bahas:

- latency
 - biaya
 - akurasi
 - bahasa Indonesia
 - bahasa Sunda
 - bahasa Jawa
-

Text To Speech

Bandingkan:

OpenAI Audio

ElevenLabs

Kokoro

Piper

Azure

Google TTS

Bahas:

- kualitas suara
- latency
- biaya
- emotion
- multilingual
- custom voice

Offline Mode

Jika internet mati.

Gunakan:

Whisper

↓

Ollama

↓

Piper / Kokoro

Jelaskan flow offline.

Language Support

Support:

Indonesia

English

Sunda

Jawa

Automatic Language Detection

Code Switching

Preferred Language

Voice Memory

Support:

Conversation Memory

Session Memory

Summary Memory

Preference Memory

Conversation Experience

AI harus mampu:

- berbicara natural
 - memahami interupsi
 - menjawab singkat ketika dibutuhkan
 - menjawab panjang ketika diminta
 - mempertahankan konteks
 - mengikuti bahasa pengguna
 - mendeteksi pergantian bahasa
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Backend Architecture

Jelaskan:

Voice Gateway

WebSocket

Conversation Manager

Voice Session

AI Router

Provider Factory

Voice Provider

Memory

Analytics

Monitoring

Mobile Architecture

iOS

AVAudioEngine

AVAudioSession

Audio Queue

Playback Queue

WebSocket

Android

AudioRecord

AudioTrack

Audio Focus

WebSocket

Monitoring

Dashboard harus menampilkan:

Current Provider

Current Model

Voice Latency

Speech Latency

Generation Time

TTFB

Streaming Delay

Playback Delay

Token Usage

Provider Health

Retry Count

Fallback Count

Turn ID

Session ID

RTT

API

WebSocket

Voice Start

Voice Chunk

Voice Stop

Voice Response

Voice Cancel

Voice Ping

Voice Pong

Health

Settings

Folder Structure

Backend

Voice Module

AI Module

Provider Module

Router Module

Memory Module

Audio Module

Monitoring Module

Acceptance Criteria

Target:

First Audio < 2 detik

Streaming Delay < 100 ms

RTT < 50 ms

Voice Interruption < 300 ms

Fallback otomatis tanpa memutus sesi

Offline mode dapat berjalan menggunakan Ollama

Support minimal Bahasa Indonesia dan Inggris

Future

Wake Word

Voice Clone

Emotion Voice

Speaker Identification

Multi Speaker Detection

Conversation Summary

Meeting Mode

Realtime Translation

AI Agent

Tool Calling

Computer Use

Gunakan diagram ASCII untuk seluruh arsitektur, sequence diagram, state machine, deployment diagram, dan class diagram.

Berikan rekomendasi teknologi terbaik beserta alasan pemilihannya untuk setiap komponen agar sistem siap digunakan pada skala production.